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1VE23CS082

Water Conservation And Rainwater Harvesting [BCV654A]

1] What is monsoon low, Describe how does it influence monsoon?

=> A monsoon low is a low-pressure system that forms over the Indian subcontinent during summer due to intense heating of land. It is a Key driver of the Indian monsoon system.

Influence on Monsoon:-

i) Wind Direction: Air moves from high pressure (ocean) to low pressure (land), bringing moisture-laden winds, southwest monsoon.

ii) Rainfall Trigger: The low-pressure zone pulls moist winds inland, leading to heavy rainfall.

iii) Monsoon Depressions: Strong monsoon lows intensify into depressions, causing widespread rainfall across India.

iv) Seasonal Control: The strength & position of monsoon low determine onset, intensity, and distribution of rainfall.

v) Withdrawal: Weakening of low pressure leads to monsoon withdrawal.

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2] Discuss the consequences of reduced surface water flow on aquatic ecosysteme and biodiversity?

=> Reduced surface water flow refers to decreased water in rivers, lakes and streams.

Effects:-

i) Habitat loss: Aquatic species lose living space, decline in fish population.

ii) Temperature Rise: Less water heats faster, affects oxygen levels.

iii) Reduced Dissolved Oxygen: Leads to fish deaths and biodiversity loss.

iv) Pollution concentration: Pollutants become more concentrated.

v) Disruption of Food Chain: Affects plankton, impacts entire ecosystem.

vi) Migration Issues: Fish cannot migrate for breeding.

vii) Wetland Degradation: Wetlands dry up, loos of biodiversity.

3] Brief the effect of urbanization on natural water cycle.

=> Urbanization significantly alters the hydrological cycle.

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Effects :-

i) Reduced Infiltration : Concrete surfaces prevent ground water recharge.

ii) Increased Runoff : Leads to floods in cities.

iii) Decreased Evapotranspiration : Less vegetation reduces moisture return to atmosphere.

iv) Ground water Depletion : Excess extraction lowers water table.

v) Water pollution : Industrial and domestic waste contaminate water bodies.

vi) Altered Drainage Patterns : Natural streams are modified or blocked.

Urbanization disrupts the natural cycle, causing water scarcity and urban flooding.

4] What is the hydrological cycle, and why is it important for maintaining life on Earth?

=> The hydrological cycle is the continuous movement of water between the atmosphere, land, and oceans.

Main Process :-

- Evaporation : Water changes into vapor.
- Condensation : Vapor forms clouds.
- Precipitation : Rainfall, snow fall.

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- Infiltration : Water seeps into ground.
- Runoff : Water flows over land.

Importance :-

- Maintains global water balance.
- Supports agriculture and ecosystem.
- Recharges groundwater.
- Regulates climate and temperature.
- Natural purification of water.
- Maintains river and lake systems.

The hydrological cycle is essential for sustaining life and ecological balance on Earth.

5] Explain the key factors contributing to the uneven distribution of water resources in Karnataka.

=> Water distribution in Karnataka varies due to natural and human factors.

Natural Factors :-

i) Rainfall variation : Western Ghats receive >3000 mm, while interior regions receive <700 mm.

ii) Topography : Mountains vs plateau affect runoff and storage.

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iii) River Basins: Rivers like Krishna and Cauvery are unevenly distributed.

iv) Soil Type: Affects water retention.

Human Factors :-

v) Over-extraction of groundwater.

vi) Poor water management practice.

vii) Unequal infrastructure development.

6] Brief the key indicators used to identify the withdrawal of monsoon rains.

=> Monsoon withdrawal marks the end of rainy season.

Indicators :-

- Reduction in rainfall activity.
- Increase in daytime temperature.
- Clear skies and reduced cloud cover.
- Decrease in humidity.
- Shift in wind direction from southwest to northeast.
- Weakening of low-pressure systems.

These indicators signify transition from monsoon to dry season.

7] How does Aquifer function in storing and supplying groundwater?

=> i) Storage of groundwater - Pores and voids store large quantities of water.

ii) Supply of groundwater - Wells and borewells extract water from aquifers.

iii) Regulation of groundwater flow - Controls underwater movement.

iv) Natural filtration - Soil layers filter impurities.

v) Drought buffer - Supplies water during dry seasons.

Aquifers act as a natural underground reservoirs, storing rainfall - recharged water and supplying it through wells and springs for dinking, irrigation, and industrial purposes.

8] Explain the different classes of rainwater harvesting techniques.

=> i) Individual / Household Systems : Small-scale systems and used in houses.

· Eg:- Rooftop rainwater harvesting
Rain barrels.

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ii) Community Systems : Large systems serving multiple households and are used in villages and towns.
· Eg :- Community tanks, check dams.

iii) Agricultural Systems : Used for irrigation in farms.
· Eg :- Farm ponds, percolation tanks.

iv) Industrial Systems : Used for industrial water supply.
· Eg :- Industrial storage reservoirs, Recycling systems.

9] Explain the feasibility of implementing rooftop rainwater harvesting in rural households.

=> Rooftop rainwater harvesting is a system where rainwater falling on roofs is collected, filtered and stored for domestic use.

· Components : Catchment (roof), Gutters, Downpipes, filter unit, Storage tank.

· Feasibility factors : Availability of rainfall, Roof area, cost of installation, Maintenance requirement, storage capacity.

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Advantages:

- Provides drinking water
- Reduces dependency on borewells
- Simple & low-cost technology.
- Supports agriculture.

Limitations:

- Seasonal rainfall dependency
- Storage tank requirement
- Regular maintenance needed.

10] Discuss the major sources of water pollution in rural and urban areas.

=> Rural sources :-

- Agriculture runoff (fertilizers, pesticides)
- Animal waste
- Open defection
- Poor sanitation
- Soil erosion

Urban Sources :-

- Industrial effluents
- Domestic sewage
- Solid waste dumping
- Oil and chemical spills
- Urban stormwater runoff.

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Effects :-

- Contamination of drinking water
- Spread of diseases
- Damage to aquatic ecosystems
- Reduced water usability.

Control Measures :-

- Wastewater treatment
- Pollution control regulations
- Public awareness
- Sustainable agriculture.

11] Explain the advantages of contour trenching towards water conservation in hilly regions.

=> Contour trenching is a soil and water conservation technique where trenches are dug along contour line on hill slopes to capture run off water.

Working :-

- Trenches constructed along contour line.
- Rainwater gets trapped.
- Water slowly infiltrates into soil.

Advantages:-
- Reduces soil erosion
- Improves soil moisture
- Increases groundwater recharge
- Enhances vegetation growth
- Prevents landslides.

Applications:-
- Hill agriculture
- Forest conservation
- Watershed management.

12] Describe the environmental and economic benefits of rainwater harvesting.

=> Environmental Benefits:-
- Groundwater recharge
- Reduced soil erosion
- Flood control
- Improved water quality
- Reduced water pollution

Rainwater harvesting helps restore natural water cycles and sustain ecosystems.

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Economic Benefits :-

- Reduces water bills
- Low installation cost
- Reduces irrigation expenses
- Increases agricultural productivity
- Decreases infrastructure cost for water supply.

Social Benefits:-

- Improves drinking water availability
- Supports rural livelihoods
- Enhances water security.

13] Explain the effect of soil type & permeability on groundwater recharge rates.

=> Groundwater recharge refers to the process by which water infiltrates the soil surface and replenishes underground aquifers. Soil type and permeability play a crucial role in determining recharge rates.

Effect of soil Type:-

i> Sandy Soil:

- Large pore spaces

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- High infitration rate
- Allows rapid groundwater recharge.
- Eg:- Desert regions often have high infiltration.

ii) Loamy Soil:
- Balanced mixture of sand, silt & clay
- Moderate infiltration rate
- Good groundwater recharge.

iii) Clay soil:
- very small pore spaces
- Low permeability
- Water infiltration is slow
- Groundwater recharge is very low.

14] How the traditional knowledge be integrated with modern technology for sustainable water management?

=> Traditional water management systems have been used in India for centuries. Integrating them with modern technologies can improve water conservation and sustainability.

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Methods of Integration:-

- Revival of Traditional Structures
- GIS and Remote sensing
- Improved Rainwater Harvesting Systems
- Smart water management.
- Community Participation
- AI Recharge Techniques

Benefits:-

- Sustainable water supply
- Increased groundwater levels
- Reduced water scarcity
- Climate resilience.